



Data structure and Programming II

TP08: Queue data structure

- Objective:
 - To practice create queue data structure that is able to store unlimited data and data is stored via FIFO method
 - Apply C++ programming
- Individual work
- Submit to Moodle (do not zip)
 - A pdf report (cover, exercise and solution with screenshot)
 - Source codes

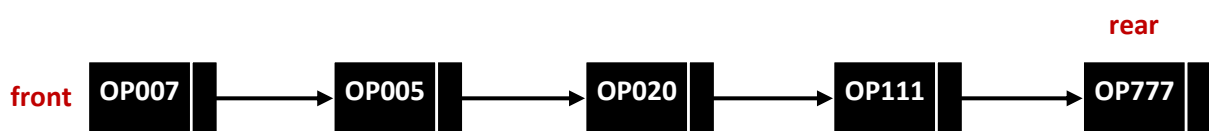
Deadline:
6 days

1. Answer the following questions:

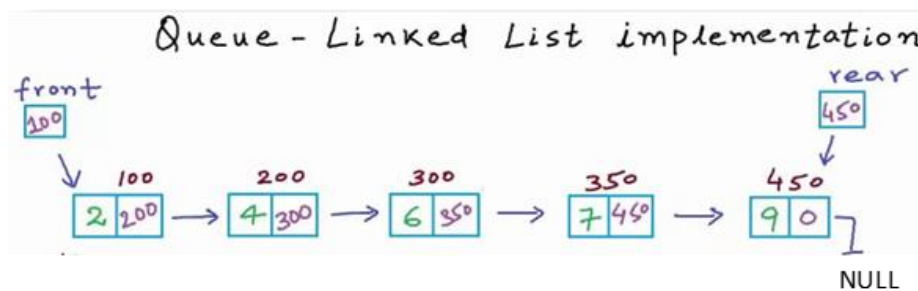
- a. What is Queue?
- b. What are the variables and functions used in Queue data structure?
- c. Give the main differences between using Linked list and using Queue.

2. Define two structures such as Element and Queue that can store queue of person. Provide at least 7 attributes (properties) of a person.

3. Write a program that can implement the Queue data structure as linked list. Create a function to initialize an empty queue. Let's assume that this queue stores ID of task operations.



4. Create an implementation of Queue as Linked list. Functions in queues are: create an empty queue, enqueue, dequeue, search data in queue, display queue. Make it as a header file (MyQueue.h) and create a main program to test the header. **Hint:** You can reuse the Single Linked List code and rename the operations and variables to the terms used by Queue.



5. Create a queue that can schedule a set of priority tasks. Each task is just to display a message on the screen. Table 1 shows about tasks and their descriptions (used for displaying message). Suppose that, the queue contains Task4, Task1, Task3, and Task2 (Task 4 is the first task in the queue, Task 1 is the second task in the queue, so on and so forth).

Write a program to ask user “*Do you want to activate a task in your queue now?*”. When the user say ‘yes’, activate the task by calling to **dequeue** function of the queue. Keep asking user until there is no task remained in the queue.

Table 1: List of tasks and their descriptions

Task	Description
1	Doing task 1 DONE!
2	Doing task 2 DONE!
3	Doing task 3 DONE!
4	Doing task 4 DONE!

6. At a restaurant, customers are waiting to be served. Create a queue as singly linked list that can store customer. Each customer has name, sex and phone number.

The program provides a menu of choices that a user can use to do some operations on the queue. The operations are:

- A. *enqueue* a customer (add a customer to queue),
- B. *dequeue* a customer (remove a customer from queue),
- C. display all customers' info in the queue and
- D. exit the program.

Each time the function *dequeue* is called, display this message "Serving customer, [*name*]!".

The program keeps running until the user choose choice number 4 to exit.

Add data to queue

Display data in queue

```
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
Enter a choice: 1
Enter a student's name: Bob
Enter a student's sex: M
Enter a student's phone number: 012123445
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
Enter a choice: 1
Enter a student's name: Jack
Enter a student's sex: M
Enter a student's phone number: 070897809
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
Enter a choice: 3
1 Bob M 012123445
2 Jack M 070897809
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
Enter a choice: 3
```

```
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
Enter a choice: 3
1 Bob M 012123445
2 Jack M 070897809
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
Enter a choice: 2
Serving customer, Bob!
*****
Welcome to the MK Restaurant!
Menu:
1. Add a customer to the queue
2. Remove a customer from the queue
3. Display queue
4. Exit the program
Enter a choice: 3
```